SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

(Deemed to be University)

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DEPARTMENT OF MATHEMATICS & COMPUTER SCIENCE

(Applicable w.e.f the Academic year 2023-2024)

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B.S. (Honours with Research) in							
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Scheme of Evaluation and Instructions for Skill Enhancement Courses

This course is designed to introduce students to algorithmic thinking. It introduces the fundamental concepts in algorithm design and how to apply these concepts in order to develop algorithms to solve specific problems.

Develop the skill set to

1. Think in the algorithmic way to solve simple computational problems
2. Choose the appropriate control structures necessary to solve specific problems
3. Formulate simple algorithms for arithmetic and logical problems
4. Test and execute the programs and correct syntax and logical errors
5. Implement conditional branching, iteration and recursion
6. Decompose a problem into functions and synthesize a complete program using divide and conquer approach
7. Use arrays and a few structures to formulate algorithms

10+2 School Level Mathematics

1	Introduction to Problem Solving and Fundamental Algorithms	Introduction, Problem Solving aspect, Top-down design, Foundation in efficiency and analysis of algorithms, Concept of a Variable, Iterative computational examples	11
2	Factoring Methods and Array techniques	Square root of a number, Greatest Common Divisor, Prime numbers, Prime factors, Raising a number to a large power, Fibonacci numbers, Simple Array operations, Removal of Duplicates, Array partitioning	11
3	Merging, Sorting and Searching	Two-way merge, Sorting by selection, Sorting by exchange, Sorting by insertion, Sorting by diminishing increment, Binary Search	11

4	Text Processing and Pattern Searching	Length adjustment, Left and right justification, Keyword search	12
5	Recursive Algorithms	Recursive Quicksort, Towers of Hanoi, Sample generation, Combination generation, Permutation generation	11
		Total	56

1. R. G. Dromey, How to Solve it by Computer, Pearson (Indian Subcontinent Low price edition), 2007. Chapters: 1 to 6, 8

2. Jean-Paul Tremblay And Richard B. Bunt, Introduction to Computer Science, An Algorithmic Approach, McGraw Hill, Second International Edition, 1981.

This lab course introduces the basics of the C language to students. It would enable a student to implement algorithms for simple problems as programs in C thereby improving algorithmic thinking.

Develop the skill set to

1. Formulate the algorithms as C programs for simple problems
2. Translate given algorithms to a working and correct program
3. Ability to correct syntax errors as reported by the compiler
4. Ability to identify and correct logical errors encountered at run time
5. Ability to write iterative as well as recursive programs
6. Ability to represent data in arrays and simple structures

10+2 School Level Mathematics

1	If-else, for & while loops	Basic programs for learning syntax, Maximum of n numbers, Printing different patterns of '*' based triangles	12
2	Iterative Programs	Square root of a number, Greatest Common Divisor, Prime numbers, Prime factors, Raising a number to a large power, Fibonacci numbers, Simple Array operations, Removal of Duplicates, Array partitioning.	18
3	Merging, Sorting and Searching	Two-way merge, Sorting by selection, Sorting by exchange, Sorting by insertion, Sorting by diminishing increment, Sorting by partitioning, Binary Search.	18
4	Recursive Programs	Recursive Quicksort, Towers of Hanoi, Sample generation, Combination generation, Permutation generation	18
5	String Manipulation programs	Length of a string, String comparison, Palindrome check, Case changing, Length adjustment, Left and right justification, Keyword search.	18
		Total	84

1. Richard F. Gilberg and Behrouz A. Forouzan, Computer Science - A Structured Programming Approach using C, ISBN-13: 978-0534491321, ISBN-10: 9780534491321, 2005.
2. Dennis Ritchie, The C Programming Language, 2nd edition, Prentice Hall India Learning Private Limited, 1990

	This course is designed to introduce the process of designing data structures to solve fairly complex problems. It imparts the following. 1. Basic concepts of data structures and algorithms 2. Concept and use of lists, trees, heaps and graphs 3. Searching techniques 4. Enable students to write algorithms for solving problems with the help of fundamental data structures
	Develop the skill set to 1. Choose the appropriate data structure for solving a specific problem 2. Analyze the algorithms to determine the time and computation complexity 3. Implement different search techniques 4. Solve multiple problems using lists, trees, heaps and graphs 5. Determine the best suited data structure to solve the problems
	UCSH-101

1	Fundamentals and Linear Lists		Abstract data type, Algorithm Analysis using Big-O notation, Stack,, Queue, Linked List, Circularly linked list, Doubly linked list, Applications	10
2	Trees		Binary Tree, General Tree, Binary Search Tree, AVL Search Tree, Applications	14
3	Heaps		Basic concepts, Operations, Algorithms, Implementation, Applications	8
4	Graph		Basic concepts, Representations, Traversals, Algorithms, Applications	14
5	Searching		Linear Search and variations, Binary Search, Hashed list search, Collision resolution	10

			Total	56
	Richard F. Gilberg and Behrouz A. Forouzan, Data Structures - A Pseudocode approach with C, second edition, Cengage Learning, 2005. 3 9, 11, 13 Mark Allen Weiss, Data Structures and Algorithm Analysis in C, Second Edition, Pearson Education, 1997.			

This lab course introduces the basics of data structures using C language. It would enable a student to design & implement data structures in C to solve fairly complex problems thereby improving data structure design skills.

Develop the skill set to
1. Program basic data structures in C.
2. Implement Lists, Trees, Heaps and Graphs appropriately to solve problems
3. Implement different searching techniques

UCSH-102

1	Pointers	Simple programs for learning C Pointer Concepts, Pointers and arrays, passing pointers in functions	6
2	Linked Lists	Implementing Singly linked list, doubly linked list, circular linked list with a few applications	18
3	Trees	Implementing General Tree, Binary Tree, Binary Search Tree, AVL Trees with a few applications	24
4	Heaps and Graphs	Implementing Heaps, Adjacency list representation of graphs, breadth first traversal, depth first traversal, basic functions in graph	24
5	Search	Linear search, Binary search, hashed list search	12
		Total	84

1. Computer Science - A Structured Programming Approach using C by Behrouz Forouzan & Richard Gilberg.
2. The C Programming Language by Dennis Ritchie
3. Yashwant Kanetkar, Pointers in C.

Semester-III

This course introduces students to the fundamental concepts of computing systems. This is to familiarize students to equip them with an understanding of hardware design including logic design, basic structure, and behavior of the essential functional modules of the computer. It also gives them how the system interacts to accomplish the required processing for the needs of the user.

Develop the skill set to:

1. To know the basics of Instruction Set Architecture (ISA) and organization of Computer components.
2. Solve simple boolean algebra problems as related to designing computer logic, through combinational and sequential logic circuits.
3. Understand how computers represent and manipulate data and convert it between different number systems.
4. Understand the memory hierarchies and processor structure and functions

NA

Unit	Topic	Content	Period
1	Arithmetic and Logic	Basics: Functional Blocks in a Computer System, Number system and Computer Arithmetic, Boolean algebra, Gates and combinational logic, Sequential Logic circuits.	20
2	Central Processing Unit	Addressing Modes, Instruction Set and Instruction Execution Flow, processor structure and function	12
3	Control Unit	Hardware and Micro-program based control Unit Design	10
4	Memory	Memory Architecture,Cache,Internal memory	12
		Total	56 Hrs

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1. William Stallings, Computer organization and architecture: designing for performance, Pearson Education India, 2000.
Chapters : 3 (3.1-3.2), 4 (4.1-4.3), 5 (5.1), 7 (7.1-7.6), 9 (9.1-9.5), 10 (10.1-10.2), 11 (11.1), 12 (12.1 to 12.4), 20 (20.1-20.5)

:

1. Carl Hamacher, Zvonko Vranesic and Safwat Zaky. “Computer Organization”, McGraw Hill, 2011.
2. David A. Patterson, John L. Hennessy. “Computer organization and design: the hardware/software interface”. Elsevier, 2011.

It emphasizes conceptual understanding, quantitative reasoning, and contemporary applications by maintaining a dynamic balance among theory, applications, modeling, and drill. Students work to develop a deeper understanding of a range of mathematical topics using interactive math tutorials that allow the student to solve math problems interactively. Students will gain some confidence on how to deal with problems which may arrive in computer science in the near future.

Develop the skill set to

1. Write and interpret mathematical notation and mathematical definitions
2. Improve the proof writing skills
3. Understand the basic mathematical structures, counting principles
4. Solve linear recurrence relations

Describe and solve some real time problems using concepts of graph theory

NA

1	Logic	Propositional Logic (PL), Applications-Logic Circuits, Propositional Equivalences, Normal forms (PCNF, PDNF), Predicates and Quantifiers, Nested Quantifiers, Rules of Inference; Introduction to Proofs-Direct Proofs, Proofs by Contraposition and Contradiction.	12
2	Counting Principles	The Basics of Counting, The Pigeonhole Principle, Permutations and Combinations, Binomial Coefficients and Identities, (combinatorial approach) Generalized Permutations and Combinations, Recurrence Relations, Modeling with Recurrence Relations, Solving Linear Recurrence Relations.	12
3	Relations	Relations and Their Properties, Representing Relations, Closures of Relations, Equivalence Relations, Partial Orderings, Hasse Diagrams.	10

4	Graphs and Trees	Graphs: Graph and sub graphs, Models, Special Types of Graphs, Representing Graphs and Graph Isomorphism, Connectivity, Euler and Hamilton Paths, Shortest-Path - Dijkstra's algorithm, Planar Graphs- Euler's Formula, Graph Coloring Trees: Introduction to Trees, Properties, Tree Traversal, Spanning Trees (BFS, DFS), Minimum Spanning Trees, Prim's and Kruskal's Algorithms.	14
5	Algebraic Systems	Definitions, Examples and general Properties, Algebraic structure, Semigroups, Monoids, Groups, Subgroups, Homomorphism and Isomorphism.	8
		Total	56

í XKenneth H. Rosen, Discrete mathematics and its applications, Eight Edition, McGraw Hill, 2019

î XTremblay J.P and Manohar R, Discrete Mathematical Structures with Applications to Computer Science, TMH Pub. Co. Ltd, 2010.

1. Kolman, Busby, Ross: Discrete Mathematical Structures, Pearson education, 4th Edition.
2. T. Veerarajan: Discrete Mathematics, with Graph Theory and Combinations(2008), Tata McGraw – Hill Publishing Company Ltd, New Delhi
3. Discrete Mathematics, Dr A Singaravelu et al, Meenakshi Agency.

Semester-IV

To understand different issues involved in the design and implementation of database systems.

Develop skill set to

1. Draw ER diagrams for a given problem
2. Design relational database with normalization
3. Handle concurrency control and storage requirements of a database

UCSH-202,UCSH-301

1	Introduction	Characteristics of database approach, Purpose of Database Systems-Drawbacks of file System, data models, DBMS architecture and data independence, E-R Modeling: Entity Course Categories, entity set, attribute and key, relationships, relation Course Categories, roles and structural constraints, weak entities, enhanced E-R and object modeling, Sub Classes:, Super classes, inheritance, specialization and generalization	14
2	File Organization	Indexed sequential access files, implementation using B++ trees, hashing, hashing functions, collision resolution, extendible hashing, dynamic hashing approach implementation and performance.	14
3	Relational Data Model	Relational model concepts, relational constraints, relational algebra. SQL: SQL queries, programming using SQL EER and ER to relational Mapping: Database design using EER to relational language, The Unified Modeling	14

		language UML-features of Good Relational Design.	
4	Data Normalization	Functional dependencies, Normal form up to 3rd normal form. Concurrency Control: Transaction processing, locking techniques and associated, database recovery, security and authorization. Recovery Techniques, Database Security	14

1. R. Elmasri and SB Navathe, Fundamentals of Database Systems, Fourth Edition, Addison Wesley, 2004

1.1, 1.2, 1.3, 2.1, 2.2, 3.3, 3.4, 3.5, 4.1, 4.2, 5.1, 5.2, 6.2, 7.1, 8.4, 9.1, 10.2, 10.3, 10.4, 13.8, 14.3, 18.1, 18.2, 19.1, 23.1

1. Abraham Silberschatz, Henry Korth, S. Sudarshan, Database Systems Concepts, 4th Edition, McGraw Hill, 1997.
2. Jim Melton, Alan Simon, Understanding the new SQL: A complete Guide, Morgan Kaufmann Publishers, 1993.

To understand and implement standard tools and techniques for software development using object oriented approach.

Develop the skill set to

1. Analyze and design real world problems using OOPS concepts viz: Encapsulation, object identity, inheritance and polymorphism.
2. Handle exceptions and write generic programming
3. Write simple C++ programs using basic object oriented paradigms

UCSH-102

Unit	Topic	Content	Periods
1	Introduction	Introducing Object-Oriented Approach, complexity of the structure and design of complex systems, Relating to other paradigms (functional, data decomposition). Basic terms and ideas: Abstraction, Encapsulation, Inheritance, Polymorphism, Review of C, Difference between C and C++ - cin, cout, new, delete operators.	14
2	Classes and Objects	Encapsulation, information hiding, abstract data types, Object & classes, attributes, methods, C++ class declaration, State identity and behavior of an object, Constructors and destructors, instantiation of objects, Default parameter value, objects, dynamic memory allocation, Metaclass/abstract classes.	14
3	Inheritance and Polymorphism	Inheritance, Class hierarchy, derivation – public, private & protected, Aggregation,	14

		composition versus classification hierarchies, Polymorphism, Categorization of polymorphism techniques, Method polymorphism, Polymorphism by parameter, Operator overloading, Parametric polymorphism	
4	Generic function	Template function, function name overloading, Overriding inheritance methods, Runtime polymorphism, Multiple Inheritance. Files and Exception Handling: Persistent objects, Streams and files, Namespaces, Exception handling, Generic Classes	14
		Total	56

1 A.R.Venugopal, Rajkumar, T. Ravishanker, Mastering C++, TMH, 1997.
Chapters: 1 (1.1.-1.18), 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14, 15, 16 (16.1-16.3), 17 (17.1-17.2), 18 (18.1-18.9), 19 (19.1-19.7)

R. Lafore, "Object Oriented Programming using C++", Galgotia Publications, 2004.
2. James Rumbaugh "Object Oriented Modeling and Design" Prentice Hall Publications, 1991
3. D. Parasons, "Object Oriented Programming with C++", BPB Publication.
4. Steven C. Lawlor, "The Art of Programming Computer Science with C++", Vikas Publication.
5. Schildt Herbert, "C++: The Complete Reference", 4th Ed., Tata McGraw Hill, 1999.
6. Tony Gaddis, Watters, Muganda, "Object -Oriented Programming in C++", 3rd Ed., Wiley Dreamtech, 2004
7. Grady Booch Object oriented Analysis and Design with applications
8. S. B. Lippman & J. Lajoie, "C++ Primer", 3rd Edition, Addison Wesley, 2000.

To analyze a real world problem and code using object oriented concepts of C++ .

Develop the skill set to

1. Understand and implement object oriented paradigms viz., Encapsulation, object identity, inheritance and polymorphism.
2. Develop a mini project.

UCSH-102, UCSH-202

1	Classes and Objects	To set and get the values given by the user, calculate the area/perimeter of geometrical shapes, Using friend functions. Static members and functions to count the number of objects created. Using constructors (default, parameterized, overloaded) defined in (local, global and function) and destructors along with their order of invocation in addition / multiplication of complex numbers (passing objects as parameters)by using inline functions. matrix multiplication using dynamic objects.	22
2	Inheritance: Single, Multiple, Hierarchical, Multilevel, Hybrid.	To write sample codes with different access specifiers, visibility modes and various types of inheritance, To use constructors and destructors in all the types of inheritance to understand the order of invocation.	12
3	Polymorphism, Templates, Exception handling,	Operator overloading on: arithmetic, logical, assignment, relational, unary, subscripting, new and delete operators, using abstract classes with virtual functions. Developing sample codes for swapping,	22

	Namespaces, Files	maximizing two numbers using templates. Handling division by zero exception, creating own namespace. To read and write into a file.	
4	Mini project	Student academic/ non-academic activities, Banking systems/ Self-reliance departments/ etc., using all the concepts.	28
		Total	84

1. A.R. Venugopal, Rajkumar, T. Ravishanker “Mastering C++”, TMH, 1997.
2. Schildt Herbert, “C++: The Complete Reference”, 4th Ed., Tata McGraw Hill, 1999.

Semester-V

To give a basic idea of an OS and its functionalities. To give an in- depth understanding of CPU and its scheduling, memory management, and device management. This course covers other functionalities of OS such as file systems, virtual memory, disk request scheduling, concurrent processes, deadlocks, security, and integrity.

Develop the skill set to

1. To become familiar with the working of the general operating system.
2. To give perspectives to understand the OS services.
3. To know the working of the operating system processes management.
4. To identify and understand memory subsystems.
5. To know the basic file management system.
6. To understand the storage structure.

NA

1	Introduction	What is an Operating System, Simple Batch Systems, Multiprogrammed Batches systems, Time-Sharing Systems, Personal -computer systems, Parallel systems, Distributed Systems, Real -Time Systems, Computing Environments, Computer System operation, I/O Structure, Storage structure and Hierarchy, Operating System components and Services- System calls, System Programs, System Structure	12
2	Processes	Process Concept, Process Scheduling, Operation on Processes CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Multiple-Processor Scheduling, Process Synchronization: Background, The Critical -Section Problem, Synchronization Hardware, Semaphores, Classical Problems of Synchronization	12
3	Deadlocks	System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from	10

		Deadlock.	
4	Memory Management	Background, Logical versus Physical Address space, swapping, Contiguous allocation, Paging, Segmentation, Virtual Memory: Demand Paging, Page Replacement, Page-replacement Algorithms, Performance of Demand Paging, Allocation of Frames, Thrashing, Other Considerations.	12
5	Storage Management	File system Implementation, Directory Implementation, Allocation methods, free space management, Disk Structure, Disk Scheduling, Disk Management, Swap-Space Management, Disk Reliability	10
		Total	56

Reference: Silberschatz and Galvin, "Operating System Concepts", Wiley-India, International Student Version, 8th Ed., 2010
Chapters:
Text 1: 1, 3 (3.1, 3.2, 3.3), 5 (5.1, 5.2,5.3, 5.5), 6 (6.1, 6.2, 6.4, 6.5, 6.6), 7, 8 (8.1, 8.2, 8.3, 8.4, 8.6), 9 (9.2,9.4,9.5,9.6,9.9), 11(11.2-11.5), 12(12.2,12.4-12.6).

	1. To develop an understanding of modern network architectures from design and performance perspective. To introduce the student to the major concepts involved in wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs). 2. To provide an opportunity to do network programming To provide WLAN measurement ideas.					
Develop the skill set to 1. Explain the functions of the different layers of the OSI Protocol. 2. Draw the functional block diagram of wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs) to describe the function of each block. 3. For a given requirement (small scale) of wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs) design it based on the market available component 4. For a given problem related TCP/IP protocol developed the network programming. 5. Configure DNS DDNS, TELNET, EMAIL, File Transfer Protocol (FTP), WWW, HTTP, SNMP, Bluetooth, Firewalls using open source available software and tools.						
UCSH-401						

1	Introduction	Introduction to computer networks, evolution of computer networks and its uses, Advantages and Disadvantages of Computer Network, reference models: OSI reference Models, TCP/IP Protocol Suite Networking fundamentals: Internet, Circuit switching vs Packet switching, ISPs, Delay and Loss in Packet Switched Networks	10
2	Local Area Network	LAN Architecture, LAN topologies- Bus/ Tree LAN, Ring LAN, Star LAN, Wireless LAN, Ethernet and Fast Ethernet, Token Ring	8
3	Application layer and data link layer	Application Layer Protocols: HTTP, FTP, SMTP, DNS Data link layer design issues, Flow Control- Stop and Wait, Error Detection, Error Control, error detection and correction, data link layer protocols,	10

		sliding window protocols, example of data link protocol- HDLC	
4	Medium access layer	Channel allocation problem, multiple access protocols, Introduction to ALOHA, CSMA/CD, CSMA/CA	8
5	The network layer	Introduction, Routers, Network layer concepts, shortest path routing, flooding, distance vector routing, link state routing (without algorithms), congestion control and quality of service, internetworking, IP, Ipv4 Addressing vs Ipv6	10
6	The transport layer	The transport layer services, elements of transport protocols, TCP and UDP, Brief introduction to presentation and session layer, E-mail	10
		Total	56

Reference: 1. Data Communication & networking: Forouzan, B. A. , Vth Ed, The McGraw-Hill Companies, Inc - 2017.
2. Data And Computer Communications, by William Stallings, VII edition, Pearson Education, 2005.

Suggested Readings:
1. Andrew S. Tanenbaum, Computer Networks, Fourth Edition, Pearson Education, 2003
2. Jim Kurose, Keith Ross, A Top Down Approach Featuring the Internet , Third Edition, Pearson Education, 2004

3	Dynamic Programming	General Method, All pairs shortest path, single source shortest path, Optimal Binary Search Trees, 0/1 Knapsack, reliability design, the traveling salesperson problem.	8
4	Traversal, Searching and Back tracking Techniques.	Traversal, searching: Techniques for Binary Trees, Graphs (BFS, DFS), connected components spanning trees, Biconnected components; Back Tracking: General Method, 8-queen's problem, sum of subsets, Graph colouring, Hamiltonian cycles, Knapsack problem.	8
5	Branch and Bound, NP-Hard and NP-Complete Problems.	Branch and Bound: The general method Least cost Search, Bounding, FIFO Branch and Bound, LC Branch and Bound, 0/1 Knapsack problem: LC and FIFO Branch and Bound Solutions; NP-Hard and NP-Complete: Basic concepts, Nondeterministic algorithms. The classes NP- hard and NP-complete. Cook's theorem; NP-Hard graph problems: CDP, NCDP, CNDP, Code generation with common subexpressions, Some simplified NP-hard Problems.	6
		Total	56

Key Text:

E Horowitz, S Sahani S Rajasekaran, Fundamentals of Computer Algorithms, Second Edition, Universities Press, 2008.

Suggested Texts:

1. Alfred V. Aho and John E. Hopcraft, and Jeffrey D. Ullman, The Design Analysis of Computer Algorithms, Pearson, 1974
2. Thomas H. Cormen, Charles E. Leiserson, R.L. Rivest, Introduction to Algorithms, Prentice Hall of India Publications, New-Delhi, 2008
3. Sara Baase and Allen Van Gelder, Computer Algorithms: Introduction to Design and Analysis, Third Edition, Pearson Education (Singapore) Pvt. Ltd, New Delhi, 2000
4. Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman, The Design and Analysis of Computer Algorithms, Pearson Education (Singapore) Pvt. Ltd New Delhi. 2012

This course introduces various methods of developing web applications

Develop the skill set to

1. Create web pages using HTML code;
2. Create web applications for client/server communication
3. Create web server applications using database connectivity.

NA

1	HTML	HTML pages using all tags; Tables and Forms	18
2	CSS	Box model; various styling options	12
3	Java Scripts	script basics; functions; objects; control statements; event handling; DOM; event handling; AJAX	18
4	Servlets	Servlet creation; request/response methods; session management; database connectivity	18
5	JSP	creating server pages with different tags; database connectivity; MVC model	18
		Total	84

Web Technologies - Theory and Practice - M Srinivasan Pearson, 2012

Web resources

1. Jon Duckett, HTML and CSS: Design and Build Websites, Wiley, 1st Edition, 2011, ISBN-13: 978-1118008188
2. Jennifer Niederst Robbins, Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics, 5th Edition, O'Reilly Media, 2018, ISBN-13: 978-1491960202

Semester-VII

The main objectives are:

To learn the different set of system calls for the Linux Operating System
To understand how the Linux OS manages files, processes and memory
To implement inter-process communication using different mechanisms

To develop skill set to:

1. Assimilate the internal abstractions of any Operating System
2. Utilize the insights gained from how these abstractions were implemented and apply them in other areas of work

UCSH-102, UCSH-201, UCSH-202, UCSH-401,

Unit	Topic	Contents	Periods
í	Introduction	System Calls, Library Functions, Standard C Library, Error handling.	ò
î	File Management	Overview, File Operations (open, read, write, lseek, close), Atomicity, File Descriptors - relation to open files and duplication, File I/O variations (pread, pwrite, readv, writev), File truncation	õ
ï	Process Management	Process concept, Process Memory Layout, Virtual Memory Management, Stack Frames, Command line arguments, Environment Variables, Process - Creation, Termination, Execution and Monitoring	í ð
ð	Memory Management	Heap and Stack Memory allocation, Memory Mapping - Creation, Unmapping, File mapping, Synchronization, Anonymous Mapping	í ð
ñ	Inter-Process Communication	Signals, Pipes, FIFO, POSIX Semaphores - Named Semaphore and Semaphore Operations, POSIX Shared Memory - Creation, Usage & Removal	í ï
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Semester-VIII

to learn fundamental theory of computer science
2. to learn formal methods of computation like automata theory, formal languages & grammars, and Turing machines.

Develop skill set to:

1. understand basic properties of formal languages and grammars
2. differentiate between regular, context-free and recursively enumerable languages
3. generate the statements of a language to construct a grammar
4. generate statements given the grammar of a language
   classify machines by their power to recognize languages

UCSH-201, UCSH-603

Unit No.	**Title**	**Contents**	**No. of Periods**
1	UNIT I	Deterministic Finite Automata, Non-deterministic automata, Equivalence of NFA and DFA, Reduction in number of states in Finite Automata	9
2	UNIT II	Regular Languages and Regular Grammars: Regular expressions, Connection between regular expressions and regular languages, Regular Grammars, Closure properties of Regular languages, Pumping lemma-Non regular languages.	9
3	UNIT III	Context free grammars ,Parsing Ambiguity, Context free grammars and programming languages ,Simplification of complex grammars and their normal forms(Chomsky and Greibach Normal form.	9
4	UNIT IV	Pushdown Automata: Nondeterministic pushdown automata, deterministic pushdown automata, Pushdown automata and context free languages, Context free grammars and pushdown automata), Properties of context	9

		free languages, Pumping lemma for non-context free languages	
5	UNIT V	Hierarchy of formal languages and automata: Recursive languages and Recursively enumerable languages, Unrestricted grammars, Context sensitive grammars and languages, Chomsky hierarchy	10
6	UNIT VI	Turing machines: Standard Turing machine, Combining Turing machine for complicated tasks, Turing thesis, Variations in Turing machine, Linear Bound automata.	10
		Total	56

Key Text: Introduction To Formal Languages And Automata, by P.Linz, Narosa Pub. 1997.

REFERENCES

1. Formal Languages, by A.Salomaa , Academic Pub.1973.
2. Introduction To Formal Languages , by Gyorgy E Revesz, Mcgraw-Hill Book Co.,1985

1. UCSH-AI-Artificial Intelligence
2. UCSH-DL-Deep Learning
3. UCSH-MLO-Machine Learning Operations
4. UCSH-NLP-Natural Language Processing

- To understand an autonomous agent
- To study a wide variety of search methods that agents can employ for problem solving.

Students will be able to

1. Understand a problem from autonomous agent's point of view
2. Solve a real-world problem using various search methods to arrive at a human like decision
3. Write inference rules based on logic that can make decisions.

Discrete Mathematics, Problem Solving with Computer, Data Structures and Algorithms

1	Introduction	What is AI? The foundations of AI, The history of AI, The state of the art; Intelligent Agents: Agents and Environments, The Concept of rationality, Nature of Environment and the structure of Agent ;	10
2	Problem Solving Problems by Searching	Solving Problems by searching: Problem solving agents, Searching for solutions, Informed and Uninformed Search strategies, Heuristic Functions;	10
3	Adversarial Search	Games, Optimal Decisions in games, Alpha, beta pruning, Imperfect Real- Time Decisions; Constraint Satisfaction Problems: Defining CSP, Constraint Propagation, Backtracking and Local Search for CSP,	12

		The Structure of Problem;	
4	Knowledge, Reasoning and Planning	Logical Agents: Knowledge-Based Agents, The Wumpus World, Logic, Propositional Logic, Propositional Theorem Proving, Effective Propositional Model Checking, Effective Agent Based on Propositional Logic; Inference in the First Order Logic: Propositional vs First Order Inference, Unification and Lifting, Forward chaining, Backward Chaining, Resolution	12
5	Classical Planning	Definition, Algorithm for Planning as State space search; Planning Graphs, Other Classical Planning approaches, Analysis of Planning Approaches; Planning and Acting in the Real World: Time, Schedules and Resources, Hierarchical Planning, Planning and Acting in Non-Deterministic Domains, Multi Agent Planning;	12
		Total	56

Text:

Stuart Russell, Peter Norvig, Artificial Intelligence: A Modern Approach, Third Edition, Pearson Education, 2011

References:

1. Elaine Rich, Kevin Knight and Shiv Shankar B. Nair, Artificial Intelligence, Third Edition, Tata McGraw Hill, 2009
2. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India), 2013

Introduce the main deep learning architectures used in the industry. Develop a strong understanding of the training mechanisms. Expose students to Advanced topics like Attention based models, Deep Generative models

Through this course, students will be able to:

1. Develop deep learning solutions to any AI related problem
2. Apply different optimization techniques to train neural networks
3. Apply different dimensional reduction techniques using neural networks
4. Develop and train a CNN model to solve a real world problem based on Image data.
5. Develop and train a RNN model to solve a real world problem based on sequential data
6. Have a strong understanding of the internal workings of advanced deep learning architectures like VAEs and GANs
7. Implement several deep learning algorithms from scratch
8. Prototype solutions in Python

Machine Learning, Optimization

1	An Introduction to Neural Networks	McCulloch Pitts Neuron, Perceptrons, Perceptron Learning Algorithm and Convergence, Multilayer Perceptron (MLPs), Representation Power of MLPs, Sigmoid Neurons, Gradient Descent, FeedForward Neural Networks, Backpropagation.	8
2	Gradient Descent Strategies	Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, RMSProp and Adams	5
3	Embedding and Representation Learning	Review of Principal Component Analysis, Autoencoders and relation to PCA, Regularization in autoencoders, Denoising autoencoders, Sparse autoencoders, Contractive autoencoders	5
4	Regularization for Deep Neural Networks	Bias variance Tradeoff, $\square^2$- regularization, Early stopping, Dataset augmentation, Parameters sharing and tying, Injecting noise at input, Ensemble methods, Dropout, Greedy Layer Wise Pre-training, Better	6

		activation functions, Better weight initialization methods, Batch Normalization	
5	Convolutional Neural Networks	The basic structure of convolutional Network: Padding, Strides, ReLU Layer, Pooling, and Fully Connected Layers; Case studies of Convolutional Architectures: AlexNet, ZFNet, VGGNet, GoogLeNet, ResNet; Visualizing Convolutional Neural Networks, Guided Backpropagation.	6
6	Recurrent Neural Networks	Recurrent Neural Networks, Backpropagation through time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT, GRU, LSTMs	6
7	Advanced topics in Deep Learning	Encoder Decoder Models, Attention Mechanism, Attention over images, Variational autoencoders, Generative Adversarial Networks (GANs)	6

Key Text(s):

1.Ian Goodfellow, YoshuaBengio, Aaron Courville, Deep Learning, The MIT Press, 2016
2.Nikhil Buduma, Nicholas Locascio, Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms, O'ReillyMedia, 2017
3. Charu C. Agarwal, Neural Networks and Deep Learning: A Textbook, Springer, 2018

1.Duda, R.O., Hart, P.E., and Stork, D.G, Pattern Classification, Wiley-Interscience, Second Edition, 2001
2. Theodoridis, S. and Koutroumbas, K., Pattern Recognition. Fourth Edition, Academic Press, 2008
3. Russell, S. and Norvig, N. Artificial Intelligence: A Modern Approach, Prentice Hall Series in Artificial Intelligence. 2003
4. Bishop, C. M. Neural Networks for Pattern Recognition. Oxford University Press. 1995.
5. Hastie, T., Tibshirani, R. and Friedman, J. The Elements of Statistical Learning. Springer. 2001.
6. Koller, D. and Friedman, N. Probabilistic Graphical Models. MIT Press. 2009.

1	PyTorch Basics	Why PyTorch?, Basics of Tensors: Tensors, indexing, size, offset, slicing, dtypes, moving tensors to GPU	10

		Representation of real-world data using tensors: image data, tabular data, and text data Download datasets, Dataset class, Dataset Transformation, Normalizing the data, Create your own dataset using *dataloader*	
2	Working with pretrained models	Pretrained models: AlexNet, ResNet, Artistic Style, CycleGAN etc	4
3	The mechanics of Learning	Gathering some data, Visualizing the data, Iterating to fit the model, Normalizing inputs, pytorch autograd, pytorch.nn module	6
4	Case studies	1. Classification of images in CIFAR10 dataset 2. Classification model to detect suspected tumors Sentiment Analysis	8
		Total	28

Key Text(s):

1. Eli Stevens, Luca Antiga, and Thomas Viehmann, Deep Learning with PyTorch, Manning Publications Co., 2020

2. PyTorch documentation, Retrieved July 05, 2022 from https://pytorch.org/docs/stable/index.html

PyTorch For Deep Learning, freecodecamp.org, retrieved July 05, 2022 from https://www.youtube.com/watch?v=GIsg-ZUy0MY

Maintenance of the Machine learning models in the production environment has become one the challenges for modern organizations using technology as well as business. This course is designed to have a basic understanding of the ML model life cycle in the production environment.

Develop the skill set to

1. Understand the MLOps as a discipline in the industry
2. Learn the ML- life cycle
3. Example based learning on ML models
4. Web Applications for ML Models
ñ Containerization

Machine Learning, Python Programming

1	MLOps: What and Why	Why Now and Challenges, People of MLOps, Key MLOps Features.	12
2	MLOps: How	Developing Models, Preparing for Production, Deploying to Production, Monitoring and Feedback Loop, Model Governance.	15
3	MLOps: Examples	Consumer Credit Risk Management, Marketing Recommendation Engines,Consumption Forecast	15
		Total	42

Key Text(s):

Mark Trveil and the Dataiku Team, Introducing MLOps: How to Scale Machine Learning in the Enterprise, O'Reilly, 2020

Retrieved Jun 30, 2022 from https://itlligenze.com/uploads/5/137039/files/oreilly-ml-ops.pdf

1	Review of ML Models	Basic Scripts-PYTHON-Model Zoo	6
2	Web Application	Converting the ML into a Web App using Flask	6
3	Containerization	Docker and Docker swarm	8
4	Scaling & Orchestration	Docker and Travis CI, AWS, and Google Kubernetes	8
		Total	28

Sandeep Giri, MLOps - Complete Hands-On Guide with Case Study, Article, August 23, 2021.

Retrieved July 5, 2022 from https://cloudxlab.com/blog/mlops-machine-learning-operations-a-complete-hands-on-guide-with-case-study/

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To impart knowledge in classification of documents, retrieving and extracting information from documents, identifying important documents available as unstructured text.

Develop the skill set to:

1. Understand basic elements of NLLP
2. Apply n-gram techniques for text processing/ text Classification
3. Apply Machine Learning models for Text Classification
4. Develop Sequence Models and Language Models for real-world problems.

Machine Learning, Deep Learning

1	Introduction to NLP	Regular Expressions, Words, Corpora, Text Normalization, Edit Distance	4
2	N-gram Language Models	N-grams, Evaluating Language Models, Sampling sentences, Generalization, Smoothing, Entropy & Perplexity	4
3	Text Classification	Naive Bayes classifier, Optimizing for sentiment analysis, Naive Bayes Language Model, Naive Bayes for other text classification tasks, Evaluation, Cross-validation, Logistic Regression, Multinomial Logistic Regression, Cross-entropy loss, Gradient Descent, Regularization	7
4	Vector Semantics	Lexical Semantics, Vector Semantics, Words & Vectors, Cosine Distance, TF-IDF, Pointwise Mutual Information, Word2Vec, Visualizing embeddings, Semantic properties of embeddings, Evaluation vector models	6
5	Sequence Labeling	Part-of-Speech Tagging, Named Entity Tagging, HMM PoS tagging, Conditional Random Fields, Evaluation	5

6	Sequence Processing using Deep Learning	Feedforward Networks for NLP, Neural Language Models, RNN Language Models, Stacked and Bidirectional RNNs, LSTM, Transformers, Contextual generation and Summarization	8
7	Machine Translation and Encoder-Decoder Models	Language Divergence, Encoder-Decoder Model, Encoder-Decoder with RNN, Attention, Beam Search, Encode-Decoder with Transformers, Bidirectional Transformer Encoders, Transfer Learning, Evaluation	8
Total			42

Key Text(s):

Dan Jurafsky and James Martin, An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition, Speech and Language Processing, Third Edition, Prentice Hall, 2022.

References:

1. Jacob Eisenstein, Introduction to Natural Language Processing, MIT Press, 2019
2. Steven Bird, Ewan Klein and Edward Loper, Natural Language Processing with Python - Analyzing Text with the Natural Language Toolkit, O'reilly First edition, 2011

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		d}š	28

1. Delip Rao and Brian McMahan, Natural Language Processing with PyTorch - Build Intelligent Language Applications Using Deep Learning, O'Reilly, First Edition. 2019.

2.Sebastian Raschka, Build a Large Language Model (From Scratch), Manning, 2024.

APPENDIX A

PRASANTHI NILAYAM- 515134
Sri Sathya Sai District, Andhra Pradesh
Tel: 08555- 287239
www.sssihl.edu.in

(Effective from 2024-2025 onwards)

Department of Mathematics and Computer Science (DMACS) offers minor courses in the following Disciplines for undergraduate students with required eligibility:

1. Mathematics
2. Actuarial Science
3. Data Science

These Minor degrees or courses will be offered subject to available logistics such as faculty, timetable slots etc. Proper understanding must be sought from the course coordinator/HoD before choosing to do a minor course in a particular semester.

A candidate who chooses to do a minor has to earn a minimum of 12 credits in 3 years of B.S program or 16 credits in 4 years of B.S Hons/B.S Hons with Research program. It is to be noted that, the minimum credits mentioned above must be earned from the courses which are of difficulty level 300 or above. A student majoring in a discipline cannot obtain a minor in the same discipline.

A candidate can choose to earn up to 12 minor credits from online MOOCs Courses.

1. Mathematics Minor

The following courses are offered for Mathematics as minor.

100 Differential Calculus
100 Integral Calculus
200 Linear Algebra -I/300 Complex Analysis
200 Algebraic Structures I/300 Methods of Differential Equations
300 Optimization Techniques
300 Numerical Analysis
400 Algebraic Structures -II
400 Mathematical Modelling

Note: For detailed syllabus, consult the syllabus document for the program *BSc Mathematics/BSc(Hons) Mathematics/BSc(Hons) Mathematics with Research*.

2. Actuarial Science Minor

The following list of courses offered for Computer Science as minor

101 Actuarial Probability and Statistics
102 Actuarial Mathematics
301 Actuarial Applied Statistical Methods
302 Business Economics with Actuarial Applications
303 Business Accounting & Finance with Actuarial Applications
401 Regulation and Financial reporting with Actuarial Application

Note: For a detailed syllabus, consult the syllabus document for the program *BSc Mathematics/BSc(Hons) Mathematics/BSc(Hons) Mathematics with Research*.

3. Data Science Minor

The following list of courses offered for Data Science as minor

Calculus provides the tools and methods for analyzing and solving problems of change and motion quantitatively. It introduces mathematical operations of differentiation and integration and develops concepts of central importance in mathematical sciences.

í X Develop the skill set to compute the slope of tangent at a point for given curve
î X write a differentiable function into a series
ï X find the maxima and minima of function in both one and two variables
ð Xcompute the area under the curve for a given surface
ñ Xperform different operators on vectors like gradient, divergence and curl
ò compute the volume of a given solid.

10 + 2 Mathematics

1	Preliminaries	Definitions and Examples of Types of sets, real-valued functions, domain, range of functions, continuous and differentiable functions, neighbourhood, intervals, limit points, boundary points, interior points, open set, closed set. Introduction to Sequences-Limit, convergence, divergence, oscillation. Introduction to Series - Convergence, divergence and oscillation, limit comparison and D'Alembert's ratio tests.	10
2	Differential Calculus	Successive Differentiation: Higher-order derivatives, Leibnitz theorem. Functions of two or more variables, partial derivatives, Euler's theorem on homogeneous functions, Total derivative, Errors and approximation, Jacobians and properties. Expansion of two variable functions using Taylor's series. Maxima and Minima of functions of two or more variables, - Lagrange's method of undetermined multipliers.	14
3	Integral Calculus	Multiple Integrals: Double and triple integrals, Change of order of Integration, Transformation of integrals from Cartesian to polar coordinates. Applications: Calculating area, volume, surface area using integrals. Beta and Gamma Functions: Definitions, properties, and relation between Beta and Gamma functions.	14
4	Solid Geometry	Equation of plane, Straight line, Intersection of lines, Shortest distance between two lines, Intersection of planes, Sphere, Equation of tangent plane, Quadratic surfaces.	10

5	Vector Calculus	Differentiation of vectors, Curves in space, Vector operator del, Gradient, Divergence and Curl.	8
			56

Dr. B.S.Grewal, Higher Engineering Mathematics, 44th Edition, Khanna Publications, 2021,

Chapters 3(3.11 to 3.18, 3.21), 4(4.1 to 4.2), 5 (5.1-5.5,5.7,5.9- 5.12), 7 (7.1 to 7.7, 7.14 to 7.16), 8 (8.1 to 8.2, 8.4 to 8.9) 9(9.1-9.6,9.8,9.9)

1 Calculus, George B. Thomas, et al, 12^{th} Edition, 2009
2. Analytical Solid Geometry by Shanti Narayana, Dr.P.K. Mittal S Chand Publishing
3. Calculus and Analytical Geometry by G.B. Thomas and R.L. Finney.

Probability and Statistics is the branch of mathematics that deals with modeling uncertainty. It is important because of its direct application in areas such as computer science, finance, and telecommunications. It forms the fundamental basis for many other areas in the mathematical sciences including modern optimization methods and risk modeling. This course provides students with a formal treatment of probability and statistics.

- íXDevelop the skill set to solve real world problems based on axioms of probability, conditional probability and Bayes' theorem
- îXunderstand the several discrete and continuous random variables
- ïXcompute covariance and correlation of two or more than two dimensional data identify statistical distributions by observing real data sets
- ðXcompute the moment-generating function for a given random variable
- ñXplot histograms, Box-plot for a given univariate data
- òXplot scatter diagram for a bi-variate data

This course can be taken after UDSC-101 Calculus

1	Elements of Probability	Sample Space, Events, counting Techniques, Axioms of Probability, Addition Rules, Multiplication Rules, Conditional Probability, Independence, Bayes' Formula and Random Variables	12
2	Discrete Random Variables	Probability Distributions and Probability Mass Functions, Cumulative Distribution Functions, Mean and Variance of a Discrete Random variable Special Discrete Random variables: Uniform, Binomial, Geometric, Negative Binomial, Hypergeometric and Poisson Distributions	12
3	Continuous Random Variables	Probability Distributions and Probability Mass Functions, Cumulative Distribution Functions, Mean and Variance of a Discrete Random variable Special Continuous Random variables: Uniform, Exponential, Normal, Erlang, Gamma, Weibull, Lognormal and Beta Distributions. Normal Approximation to the Binomial and Poisson Distributions	14

4	Joint probability Distributions	Joint distribution of Two and more than two Random Variables, Conditional Probability Distributions and Independence, Covariance and Correlation, Common Joint Distributions, Linear Functions and General Functions of Random Variables, Moment-Generating Functions	12
5	Descriptive Statistics	Numerical Summaries of Data, Stem-and-Leaf Diagrams, Frequency Distributions and Histograms, Box Plots, Scatter Diagrams, Probability Plots.	6
		Total	56

Montgomery Douglas C., Runger George C., Applied Statistics and Probability for Engineers, Seventh Edition, Wiley, 2018, Chapters: 2 - 6.

- Sheldon M Ross, Introduction to Probability and Statistics for Engineers and Scientists, 6th Edition, Elsevier Academic Press, 2020
- Irwin Miller Marylees Miller, John E. Freund's Mathematical Statistics with Applications, 8th Edition, Pearson Publications

: The main purpose of Statistical Inference is to make an inference about a population based on information contained in a random sample selected from that population, and to provide an associated measure of goodness for the inference. Another purpose is hypothesis testing for mean and variance of a normal distribution.

:

- í X Develop the skill set to Estimate both point and interval estimations for a given parameters based on a sample from a normal population
- î X Analyze the estimated parameter
- ï X Find an estimator for population proportion based on sample data
- ð XDevelop testing hypothesis for a given real world problem
- ñ XApply different statistical tests based on data
- ò XModel the relationship between a scalar response and one or more explanatory variables
- **7.** Find an appropriate regression model for a given data

Probability and Statistics, Calculus

1	Point Estimation	Sampling Distributions and the Central Limit Theorem, Point Estimation: Unbiased Estimators, Variance of a Point Estimator, and Mean squared Error of an Estimator, and Methods of Point Estimation: Method of Moments, Maximum Likelihood Estimators, Bayesian Estimation of Parameters;	10
2	Interval Estimation for a Single Sample	Confidence Intervals on mean of a Normal Distribution, Confidence Interval on variance of a Normal Distribution, Large Sample Confidence Interval for a Population Proportion, Bootstrap Confidence Interval, Tolerance and Prediction Intervals	9
3	Testing Hypotheses for a Single Sample	Hypothesis Testing, Tests on the Mean and variance of a Normal Distribution, Tests on a Population Proportion, Testing for Goodness of Fit and Contingency Table Tests	10
4	Statistical Inference for Two samples	Inference on the Difference in Means of two Normal Distributions, A Nonparametric Test for the Difference in Two Means, Paired t-test, Inference on the variance	12

		of Two Normal Distributions, Inference on Two Population Proportions	
5	Simple Linear Regression	Simple Linear Regression, Least squares estimators,Hypothesis tests and Confidence Intervals in Single Linear Regression, Adequacy of the Regression Model: Residual Analysis, Coefficient of Determination, Regression on Transformed Variables, Logistic Regression, Polynomial Regression Models	15
		Total	56

Montgomery Douglas C., Runger George C., Applied Statistics and Probability for Engineers, Seventh Edition, Wiley, 2018
Chapters: 7 -9 (up to 9.8), 10, 11

- Sheldon M Ross, Introduction to Probability and Statistics for Engineers and Scientists, Sixth Edition, Elsevier Academic Press, 2020
- Irwin Miller, Marylees Miller, John E. Freund's Mathematical Statistics with Applications, Eighth Edition, Pearson Publications

This course provides an introduction to Linear algebra and its applications to other disciplines.

By the end of the course, students will be able to:

1. Understand the use of linear algebra in modeling concepts.
2. Perform matrix calculations effectively.
3. Describe the concepts of vector spaces and their subspaces through various examples.
4. Differentiate between linearly independent and dependent sets and determine the basis for finite-dimensional vector spaces.
5. Analyze linear transformations between vector spaces, discern their properties using the image and kernel of the transformation.
6. Express linear transformations in matrix form and discuss their properties.
7. Understand the use of SVD.
8. Explain characteristic polynomials and comprehend the concepts of eigenvalues and eigenvectors in the context of endomorphisms.

10+2 Mathematics, UDSC-101

1	Vector Spaces	Vectors - Linear Combinations, Lengths, Dot products. Matrices-Linear Equations, Elimination method, Rules of Matrix operations, Inverse, LU factorization, Transposes and permutations.	8
2	Fundamental Spaces	Vector space-Spaces of vectors, Null space, Complete solution to nonhomogeneous equation, Independence, Basis, Dimension. Subspaces - Dimension of Four subspaces, Orthogonality, Projection, Orthonormal bases, Gram Schmidt orthogonalisation	8
3	Determinants	Determinants-Properties, Permutations and Cofactors, Cramer's rule, Inverse, Volume. Eigenvalues and Eigenvectors - Eigenvalues, Diagonalisation, System of differential equations, Symmetric matrices, Positive Definite matrices	8
4	SVD	Singular Value Decomposition - Image processing, Bases and matrices, Principal Component Analysis, Geometry of SVD	10

5	Linear Transformations	Linear Transformations - Introduction, Matrix of Linear Transformation, Search for a good basis	8
		Total	42

Gilbert Strang, Introduction to Linear Algebra, 5th Edition, Wellesley-Cambridge Press, 2016
: 1 (1.1-1.3), 2 (2.1-2.7), 3 (3.1-3.5), 4 (4.1-4.4), 5 (5.1-5.3), 6 (6.1-6.5), 7 (7.1-7.4)
https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/
Retrieved on Jan 31, 2023

- Stephen Boyd & Lieven Vandenberghe, Introduction to Applied Linear Algebra - Vectors, Matrices and Least Squares, Cambridge University Press, 2018
- Philip N. Klein, Coding the Matrix - Linear Algebra through Applications to Computer Science, Newtonian Press, 2013

1	Python for Linear Algebra	The following programs are to be implemented in basic python - Dot Product of Vectors, Matrix Multiplication with basic python - Determinant of a Matrix, Matrix Inverse with basic python - Rank of a Matrix with basic python - Gaussian Elimination with basic python	12
2	Numpy and its usage in Linear Algebra	The following are to be implemented using numpy - Eigenvalues, Eigenvectors, SVD - Eigenfaces - Linear Regression - Matrix Exponential	16
		Total	28

Leo Chin, Tanmay Dutta, NumPy Essentials, Packt Publishing, 2016

Umit Mert Cakmak, Mert Cuhadaroglu, Mastering numerical computing with Numpy, Packt Publishing, 2018

To modeling and discussion of documented real-world applications. Study of mathematical programming algorithms for optimization. Apply the mathematical results and numerical techniques of optimization theory to concrete Engineering problems.

Develop the skill set to

- í X discovery, study and solve optimization problems;
- î X Investigate, study, develop, organize and promote innovative solutions for various applications.
- ï X Apply mathematical skills to model optimization problems.
- ð XCritically analyze theoretical principles and choose the relevant optimization techniques for a specific problem.
- ñ XExposed to solutions for various types of optimization problems and will be enabled to adopt them for the situation at hand.

10+2 Mathematics, Python programming, UDSC-101 Calculus

	Introduction	Mathematical Review – Lines – Hyperplanes and linear varieties – convex sets – neighborhoods – polytopes and polyhedra – derivative matrix – level sets and gradients- Taylor's series	6
	Unconstrained Optimization	Unconstrained Optimization – conditions for local minimizers – one-dimensional search methods including golden section, Fibonacci, bisections and Newton's method – Gradient methods including steepest descent methods, Newton's method and its analysis.	8
	Linear Programing	Modeling with Linear Programming The Simplex method and Duality	8
	Non-Linear Constrained Optimization	Non-linear constrained optimization – problems with equality constraints – Lagrange condition – second-order condition – minimizing quadratics subject to equality constraints – problems with inequality constraints – Karush-Kuhn-Tucker conditions – second-order conditions.	8

	Convex Optimization	Convex Optimization Problems - Convex Functions - Convex Optimization Problems	6
	Machine Learning view for Optimizatio	The minutiae of Gradient Descent, Properties of Optimization in ML, Computing Gradient wrt vectors, Linear Regression and Optimization models for Binary Targets	6
		Total	42

í X L.R. Foulds, Optimization Techniques, Springer, Utm, 1981
î X http://freevideolectures.com/Course/3072/Numerical-Optimization
ï X Hamdy A. Taha, Operations Research, Pearson Publication, 9th edition, (2014).
ð X Boyd, Stephen, and Lieven Vanderberghe, Convex Optimization. Cambridge, UK: Cambridge University Press, 2004.

1	Unconstrained Optimization	Implementation of Unconstrained Optimization using Python: Newton-Raphson method; Implementation of Gradient Ascent and Gradient Descent methods	10
2	Solving LPP	Solving LP problems using Excel solver; Sensitivity Analysis using Excel solver; Solving LP problems using *scipy.optimize.linprog*	10
3	Python Library *scipy.optimize*	Univariate function minimization; Constrained and Unconstrained minimization of multivariate scalar function; Global Optimization; Least-Square Optimization; Custom minimizers	8
		Total	28

í X SciPy Reference Guide, Retrieved July 7, 2022 from https://docs.scipy.org/doc/scipy-1.7.1/scipy-ref-1.7.1.pdf

Section 2.4 (Optimization - *scipy.optimize*)

î X Christian Hill, Learning Scientific Programming with Python, Second Edition, Cambridge University Press, 2020

ï X Edwin K. P. Chong and Stanislaw H. Zak, An Introduction to Optimization, Fourth Edition, Wiley Inter - Science Publication, John Wiley and Sons Inc., 2013.

ð XHamdy A.Taha, Operations Research- An Introduction, 10th Edition, Pearson Education - 2017

- To introduce students to the basic concepts and techniques of Machine Learning.
- To become familiar with regression methods, classification methods, Pattern Mining and clustering methods.
- To become familiar with Kernel Methods and Dimensionality reduction Techniques

Develop the skill set to

- Gain knowledge about basic concepts of Machine Learning
- Identify machine learning techniques suitable for a given problem
- Solve the problems using various machine learning techniques
- Apply Dimensionality reduction techniques

UDSC-101/UMAT-101, UDSC-102, UDSC-201, UDSC-203 or Equivalent

1	Introduction	Data Matrix, Numeric Attributes, Categorical Attributes, Graph Data, Kernel Methods, High-dimensional Data, Dimensionality Reduction, Types of Machine Learning	8
2	Regression	Linear Regression, Logistic Regression, Neural Networks, Regression Evaluation	8
3	Classification	Probabilistic Classification, Decision Tree Classifier, Linear Discriminant Analysis, Support Vector Machines, Classification Assessment.	12
4	Clustering	Representative-based Clustering, Hierarchical Clustering , Density-based Clustering, Spectral and Graph Clustering, Clustering Validation	14

5	Frequent pattern mining	Itemset Mining, Summarizing Itemsets Sequence Mining, Graph Pattern Mining.	14
		Total	56

1. Mohammed J. Zaki, Wagner Meira, Jr., Data Mining and Machine Learning: Fundamental Concepts and Algorithms, 2nd Edition, Cambridge University Press, March 2020.
2. Trevor Hastie, Robert Tibshirani, and Jerome H. Friedman, The Elements of Statistical Learning, Second Edition, Springer, 2009
   Retrieved on July 05, 2022 from https://hastie.su.domains/Papers/ESLII.pdf

4. Chrisopher M Bishop, Pattern Recognition and Machine Learning, Springer, 2006
5. Rogers and Girolami, A First Course in Machine Learning, Chapman and Hall/CRC, 2015

References:

- íXBarber, Bayesian Reasoning and Machine Learning, Cambridge University Press, 2012
- îXHal Daumé III, A Course in Machine Learning, e-Edition, 2012
- ïXMitchell, Machine Learning, McGraw Hill, 1997
- ðXCS229-Machine Learning, Stanford University, Retrieved June 30, 2022 from https://see.stanford.edu/Course/CS229

	Introduction	Review numpy, pandas and matplotlib for statistical, Probabilistic, algebraic and geometric views of data matrix Implementation of Linear algorithms, Non-Linear Algorithms and ensembling algorithms, bias-variance tradeoff, from scratch in python	12
	Python Library scikit-learn	Develop different ML algorithms using scikit-learn	16

Key Text(s):

- íXJason Brownlee, Master Machine Learning Algorithms, Discover How They Work and Implement them from Scratch, Machine Learning Mastery, eBook, 2017
  Retrieved June 30, 2022
  https://github.com/AmandaZou/master-machine-learning-algorithms
- îXAurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'reilly, 2019
- ïXScikit-learn Documentation, retrieved June 30, 2022 from scikit-learn Tutorials — scikit-learn 1.0.2 documentation

- íX To provide an overview of an exciting growing field of big data analytics.
- îX To teach the fundamental techniques and principles in achieving big data analytics with scalability and streaming capability.
- ïX To enable students to have skills that will help them to solve complex real-world problems in decision support.

Students will be able to

- íX Understand the key issues in big data management and its associated applications in intelligent business and scientific computing.
- îX Interpret business models and scientific computing paradigms, and apply software tools for big data analytics.
- ïX Achieve adequate perspectives of big data analytics in various applications like recommender systems, social media applications etc.

UDSC-302 Data Mining and Machine Learning

1	Introduction to Big Data	Big Data - Why and Where?, Characteristics of Big Data and Dimensions of Scalability, Getting value out of Big Data, Foundations for Big Data Systems and Programming	6
2	Similarity Algorithms	Near-Neighbor search, Shingling, Similarity preserving summary, Locality sensitive functions, Distance measures, Locality sensitive hashing and its applications to different distance measures, Applications of Locality sensitive hashing	12
3	Streaming Data	Stream Data model, Sampling data in a stream, Filtering streams, Counting distinct elements in a stream, Application of stream algorithms in counting.	10
4	Link Analysis	Page Rank, Computation of Page Rank, Topic sensitive page rank, Link spam.	6
5	Frequent Item sets	Market-Basket model, A-priori algorithm, Larger datasets in main memory, Limited pass algorithms, Counting frequent sets in a stream	10

6	Social Network Graphs	Clustering, Discovering of communities, Partitioning, Finding overlapping communities, Simrank, Counting Triangles, Neighborhood properties	12
		Total	56

Anand Rajaraman, Mining Massive Datasets, Stanford University Press, 2014
: 2.1 - 2.3, 3.1 - 3.7, 4.1 - 4.6, 5.1 - 5.4, 6.1 - 6.4, 10

This course gives students an insight into the basics of cloud computing along with virtualization, cloud computing is one of the fastest growing domains from a while now. It will provide the students basic understanding about cloud and virtualization along with how one can migrate over it.

Develop the skill to

- í X analyze the cloud computing setup with its vulnerabilities and applications using different architectures.
- î X design different workflows according to requirements and apply map reduce programming model.
- ï X apply and design suitable virtualization concepts, cloud resource management and design scheduling algorithms.
- ð X create combinatorial auctions for cloud resources and design scheduling algorithms for computing clouds
- ñ X assess cloud storage and security, the risks involved, its impact and develop cloud applications
- ò X broadly educate to know the impact of engineering on legal and social issues involved in addressing the security issues of cloud computing.

	Cloud Computing Overview	Origins of Cloud computing – Cloud components - Essential characteristics – On-demand self service, Broad network access, Location independent resource pooling ,Rapid elasticity , Measured service, Comparing cloud providers with traditional IT service providers, Roots of cloud computing.	3
	Cloud Insights	Architectural influences – High-performance computing, Utility and Enterprise grid computing, Cloud scenarios – Benefits: scalability ,simplicity ,vendors ,security, Limitations – Sensitive information - Application development- security level of third party - security benefits, Regularity issues: Government policies.	3

	Cloud Architecture - layers and Models	Layers in cloud architecture, Software as a Service (SaaS), features of SaaS and benefits, Platform as a Service (PaaS), features of PaaS and benefits, Infrastructure as a Service (IaaS), features of IaaS and benefits, Service providers, challenges and risks in cloud adoption. Cloud deployment model: Public clouds	10
	Cloud Simulators CloudSim and Green Cloud	Introduction to Simulator, understanding CloudSim simulator, CloudSim Architecture(User code, CloudSim, GridSim, SimJava) Understanding Working platform for CloudSim, Introduction to GreenCloud	20
	Introduction to VMWare Simulator	Basics of VMWare, advantages of VMware virtualization, using Vmware workstation, creating virtual machines-understanding virtual machines, creating a new virtual machine on local host, cloning virtual machines, virtualizing a physical machine, starting and stopping a virtual machine.	20
			56

1. Cloud computing a practical approach - Anthony T.Velte , Toby J. Velte Robert Elsenpeter, TATA McGraw- Hill , New Delhi – 2010
2. Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online - Michael Miller - Que 2008

1. Cloud computing for dummies- Judith Hurwitz , Robin Bloor , Marcia Kaufman ,Fern Halper, Wiley Publishing, Inc, 2010
2. Cloud Computing (Principles and Paradigms), Edited by Rajkumar Buyya, James Broberg, Andrzej Goscinski, John Wiley & Sons, Inc. 2011

APPENDIX B

Skill Enhancement Courses

Course Objectives: This course provides an introduction to the Python programming language. It covers the fundamentals of programming, including variables, data types, control structures, and functions. Students will learn how to write Python programs to solve simple problems and develop basic programming skills.		
	Introduction to Programming Overview of the Python programming language Setting up the development environment Running Python program	2
	int, float, complex, str, list, tuple, bytes, byte array, dictionaries Boolean type, Arithmetic operator, Relational operator, Logical or Boolean operator, Assignment Operator, Bitwise operator	8
	Conditionals: if, else, elif statements Loops: for and while loops Loop control statements: break, continue	8
	Python built in functions, Python user defined functions, Python packages functions; Defining and calling Function, The anonymous Functions	10
		28
Mark Lutz, Learning Python, 5th Edition, O'Reilly Media Publishers Think Python: How to think like a Computer Scientist by Allen Downey, 2nd edition, O'Reilly Publications: https://greenteapress.com/wp/think-python-2e/ https://www.w3schools.com/python		

The course aims to:
introduce students to advanced features of Python, such as complex data types, classes, and file handling, introduction to object-oriented Programming and data visualization

Course Outcomes: At the end of the course, the student will be able to:

- write and execute simple programs using strings, dictionaries, and tuples, and on classes and file handling in Python
- create classes and objects
- sketch graphs using appropriate Python libraries, and interpret them for a better understanding
- develop proficiency in creating and manipulating NumPy arrays.
- use mathematical and statistical operations on arrays.

	Modules and Classes: How to write a module in Python, writing various customized modules for the purpose of reuse. Also introducing the concept of object-oriented programming using classes, inheritance, and encapsulation.Understanding List,Dictionary and Tuple Operations	10
	Introduction to matplotlib (Using the pyplot submodule) figure, axes, subplot plot(), scatter(), show(), labels, legends, titles, styles, ticks, dynamically updating curves, saving graphs	6
	Introduction to numpy The numpy array – properties: size, shape, ndim, dtype, creating arrays, zeros, one(), full(), fill(), arange(), linspace(), logspace(), identity(), eye(), astype(), indexing and slicing arrays (view versus copy), important array methods- reshape(), ravel(), flatten(), hstack() and vstack(), Aggregate functions- p.sum(), np.prod(), np.mean(), np.std(), np.var(), np.min(), np.max(), np.argmin(), np.argmax()	12
		28

1. Think Python: How to think like a Computer Scientist by Allen Downey, 2nd edition, O'Reilly Publications: https://greenteapress.com/wp/think-python-2e/
2. Mark Lutz, Learning Python, 5th Edition, O'Reilly Media Publishers
3. https://www.w3resource.com/python/python-tutorial.php

To make Data Storytelling through visualization of the data

: Develop the skill set to
1. Distinguish qualitative and quantitative data
2. Make inferences out of the different plots
3. Develop story out of the data

Basic Python/R

1	Numbers that Summarize the data	Measures of Average, Measures of Variance, Measures of correlation and Measures of Ratio.	14
2	Fundamental Variations of Graphs	A Brief History of Graphs; Graphical means for Qualitative data; Visual attributes: Lines, Histograms, Bar plot, Scatter plot and Box-Plots and inferences through examples	14
3	General Design Principles for Communication	Organizing, Highlighting, Integration, Table Design, General Graph Design, Multi-Variable display;	14
4	Case Studies	Take different data sets and make data storytelling with visualization.	14
		Total	56

*For implementation: Instructor can use any visualization framework
Matplotlib/Seaborn/Plotly, ggplot2, Tableau, PowerBI, or any other

1. Stephen Few, Show me the numbers: Designing tables and graphs to enlighten, Second Edition, Analytic Press, 2012 Chapters - 1 to 11

2. Cole Nussbaumer Knaflic, Storytelling with Data, John Wiley & Sons, Inc. - 2015
3. Matplotlib Documentation, Retrieved June 30, 2022 from https://matplotlib.org/3.5.1/plot_Course Categorys/index.html
4. Seaborn Documentation, Retrieved June 30, 2022 from https://seaborn.pydata.org/introduction.ht

Course Title	Software Lab in SQL		L	T	P	C	P
Course Code	UCSH-404		0	0	4	2	56
Course Category	SEC						

Course Objectives: It introduces basic query writing using SQL and MYSQL workbench.
Course Outcome: Develop skill set to

1. Write simple DDL statements using MySQL and MySQL Workbench
2. Write simple DML statements MySQL and MySQL Workbench
3. Implement triggers, logs along with locks

Prerequisites for the Course: NA

Unit	Topic	Contents	Periods
1	SQL DDL (data Definition language)	Working on MYSQL command prompt as well as MYSQL workbench : To Create, insert ,delete, drop, alter, set, rename, operations on the databases (at least 6 tables from the databases mentioned) using with keys viz., primary, foreign and basic data Course Categories viz., varchar, int, float, date, time etc.	20
2	SQL DML (Data manipulation languages)	MYSQL command prompt as well as MYSQL workbench : Using Select, from, where for single as well as multiples tables, Using Aliases, distinct,*,Union, minus and intersect erations, Nested queries, Exists function, use of NULL, Using aggregate functions viz, max, min and avg.	20
3	Triggers	Triggers, Locks and Logs	16
		Total	56

References:

MYSQL reference manual 5.7. https://downloads.mysql.com/docs/workbench-en.pdf

The aim of the Actuarial Mathematics subject is to provide a grounding in financial mathematics and its simple applications		
At the end of the course, the student will be able to: 1. Understand and be able to perform calculations relating to present value, current value, and accumulated value. 2. Calculate present value, current value, and accumulated value for sequences of non-contingent payments. 3. Understand key concepts concerning loans and how to perform related calculations. 4. Understand key concepts concerning bonds, and how to perform related calculations 5. Understand key concepts concerning yield curves, rates of return, measures of duration and convexity, cash flow matching and immunization, and how to perform related calculations.		
	Time Value of Money a. Define and recognize the definitions of the following terms: interest rate (rate of interest), simple interest, compound interest, accumulation function, future value, current value, present value, net present value, discount factor, discount rate (rate of discount), convertible monthly, nominal rate, effective rate, inflation and real rate of interest, force of interest, equation of value. b. Given any three of interest rate, period of time, present value, and future value, calculate the remaining item using simple or compound interest. Solve time value of money equations involving variable force of interest. c. Given any one of the effective interest rate, the nominal interest rate convertible m-thly, the effective discount rate, the nominal discount rate convertible m-thly, or the force of interest, calculate any of the other items. d. Write the equation of value given a set of cash flows and an interest rate.	
	Annuities/cash flows with non-contingent payments a. Define and recognize the definitions of the following terms: annuity-immediate, annuity due, perpetuity, payable m-thly or payable continuously, level payment annuity, arithmetic increasing/decreasing annuity, geometric increasing/decreasing annuity, term of annuity. b. For each of the following types of annuity/cash flows, given sufficient information of immediate or due, present value, future value, current value,	

	interest rate, payment amount, and term of annuity, calculate any remaining item. o Level annuity, finite term. o Level perpetuity. o Non-level annuities/cash flows. Arithmetic progression, finite term and perpetuity. Geometric progression, finite term and perpetuity. Other non-level annuities/cash flows.	
	a. Define and recognize the definitions of the following terms: principal, interest, term of loan, outstanding balance, final payment (drop payment, balloon payment), amortization. b. Calculate: • The missing item, given any four of: term of loan, interest rate, payment amount, payment period, principal. • The outstanding balance at any point in time. • The amount of interest and principal repayment in a given payment. • Similar calculations to the above when refinancing is involved	
	Bonds a. Define and recognize the definitions of the following terms: price, book value, market value, amortization of premium, accumulation of discount, redemption value, par value/face value, yield rate, coupon, coupon rate, term of bond, callable/non-callable, call price, call premium, accumulated value with reinvestment of coupons. b.Given sufficient partial information about the items listed below, calculate any of the remaining items • Price, book value, market value, accumulated value with reinvestment of coupons, amortization of premium, accumulation of discount. (Note that valuation of bonds between coupon payment dates will not be covered). • Redemption value, face value, Yield rate, Coupon, coupon rate. • Term of bond, point in time that a bond has a given book value, amortization of premium, or accumulation of discount. Calculate the price of a callable bond to achieve a specified minimum yield.	

Broverman, S.A., Mathematics of Investment and Credit (Seventh Edition), 2017, ACTEX Publications, ISBN 978-1-63588-221-6

Chapter 1 (excluding 1.2.1 and 1.8), Chapter 2 (excluding 2.3.1.2, 2.4.2, 2.4.3 and 2.4.5), Chapter 3 (excluding 3.2.1, 3.2.2, 3.3, and 3.4), Chapter 4 (excluding 4.1.3), Chapter 5 (excluding 5.2, the investment year method portion of 5.3.1, and excluding all of 5.3.2, 5.3.3 and 5.3.4), Chapter 6 (excluding 6.2 and 6.4), Chapter 7 (excluding 7.1.3, 7.1.6 and 7.3)

Bowers, N. L.; Gerber, H. U.; Hickman, J, C. et al., In Actuarial Mathematics, 2nd ed. Society of Actuaries, 1997. 753 pages. ISBN: 9780938959465.

Butcher, M. V., Mathematics of compound interest, Nesbitt, C. J. Ulrich's Books, 1971. 324 pages. ISBN: 9780960300013.

Ingersoll, J. E. Rowman & Littlefield, 1987, Theory of financial decision making. 474 pages. ISBN: 9780847673599.
Kellison, S. G., The theory of interest. 3rd ed. Irwin, 2008. 463 pages. ISBN: 9780073382449.
Gerber, H. U., Life insurance mathematics, 3rd ed. Springer; Swiss Association of Actuaries, 1997. 217 pages. ISBN: 9783540622420.

APPENDIX C

Multi-Disciplinary Courses

	MDC	
10+2 any Discipline		
The objective is to give basic knowledge of the subject and familiarity with techniques to use them effectively. Knowledge of Mathematics at the Senior Secondary level will be an advantage but not necessary for offering this course. The scope of Statistics has been well recognized over the years but has very much expanded in recent years and with the impressive progress in		
At the end of the course, the student will be able to: í X Recognize and differentiate between key terms. î X Apply various types of sampling methods to data collection. ï X Create and interpret frequency tables. ð XDisplay data graphically and interpret graphs: stemplots, histograms, and box plots. ñ XRecognize, describe, and calculate the measures of location of data: quartiles and percentiles. ò XRecognize, describe, and calculate the measures of the center of data: mean, median, and mode. ó XRecognize, describe, and calculate the measures of the spread of data: variance, standard deviation, and range.		
	What is statistics, Uses and misuses of statistics, Introduction to basic terms, Measurability and variability, Relationship between probability and statistics?	
	Data collection, Measures of central tendency, Measures of Dispersion, Graphic Presentation and Measures of Position.	

	Frequency distributions, Graphic presentations, Calculations using frequency distribution, Interpreting and understanding Standard Deviation, The art of Statistical Deception.	
	Bivariate Data, Linear Correlation, Linear Regression	

Robert R. Johnson, Elementary Statistics, Duxbury Press, North Scituate, Massachusetts, 1976, 2nd edition

1,2,3, 4

1. Mario F. Triola, Elementary Statistics, Addison-Wesley, 7th Edition, 1997.
   Robert N. Goldman, Joel S. Weinberg, Statistics An Introduction, Prentice-Hall Publication 1985.

The course aims to:

- Give a basic understanding of working with numbers which include Modular Arithmetic, the Principle of Induction, and the usage of prime numbers.
- Give a basic understanding of Geometry, Euclid's elements, conic sections, non-Euclidean geometry, and higher dimensions
- Give a basic understanding of Probability, experiments, and chance

At the end of the course, the student will be able to:

- Start any abstract course on Number Theory
- Read courses on Applied and Theoretical Probability.

	You can count on them;These days, children do it before they are five;A token advance ; Symbolic Progress; For a long time, it was all Greek;A fatal law is discovered; Here's looking at Euclid; Numbers in Prime condition; Prime order;The child Genius; Gauss's clock Arithmetic;The Great Amateur;Taking the prime test;Keeping Secrets; Easy to Guess, Hard to prove;Fermat's Last Theorem;The Fermat Saga Begins;The Domino Effect	18
	Everyone is a geometer;Measuring the Earth;Things that Euclid missed;Euclid in his elements;Geometry as a theory of Everything;Slicing through Cones;A fly on the ceilin Squaring the circle and other impossibilities;The startling discovery of non-Euclidean geometries;The geometry of Renaissance artists;Beyond the third dimension	18
	Who gets Heaven?;Figuring the odds;The geometric patterns of chance;In Mathematics we trust;The Amazing Bernoulli;Fear of Flying;Ringing in the bell curve;The Reverend Bayes;Enter the Average Man;The abstract Patterns of chance;Using Mathematics to choose your options	20

1. Keith Devlin, The Language of Mathematics (Making the invisible visible), W H Freeman and Company, 2000, 1st edition	Numbers, Geometry and Probability	Chapter: 1, 4, 7

Suggested Readings:

1. Richard Courant and Herbert Robbins, what is Mathematics? Oxford University Press, 1941, 2nd Edition